# Fair Division of Indivisible Goods

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by

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# CERTIFICATE

This is to certify that the work contained in the project entitled ***"Fair Division of Indivisible Goods"*** is a bonafide work of ***Saurabh kumar (Roll No. 112402005)***, carried out in the Department of Computer Science and Engineering, ***Indian Institute of Technology Palakkad*** under my guidance and that it has not been submitted elsewhere for a degree.

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***Saurabh kumar***

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# Chapter 1

# Introduction

## 1.1 Introduction

The problem of allocating resources fairly among agents with heterogeneous preferences is a central theme in economics, mathematics and also in computer science. The goal of *fair division* is to distribute goods in a manner that satisfies a well defined notion of fairness even when individuals evaluate items differently. This question is not just theoretical but also real-world platforms such as CourseMatch for course allocation at the Wharton School [1] and public-facing systems like Spliddit and Fair Outcomes [23] highlight the growing practical need for algorithms that compute fair assignments in complex environments.

The roots of fair division trace back to classical work by Steinhaus, Banach, and Knaster, who initiated the formal study of dividing *divisible* resourcesmost famously, cake cutting [14]. In these divisible settings, strong fairness criteria such as proportionality and envy-freeness can often be achieved, and major progress has shown that envy-free cake-cutting procedures can be implemented in finitely many steps [15].

On the other hand when resources are *indivisible* such as discrete objects, jobs or housing assignments fair division becomes much more challenging. Classical fairness guaranties that it might fail entirely as items cannot be subdivided without losing their value. This has motivated the use of randomized mechanisms [21] and the development of relaxed or approximate fairness notions like EF1 (envy-freeness up to one good) and EFX (envy-freeness up to any good) as well as MMS (maximin share fairness) [22, 1, 2]. A significant amount of research on the existence of fair allocations and effective algorithms for calculating them has been motivated by these ideas.

Among these ideas *EFX* is one of the most robust fairness guarantees that is still relevant for indivisible items. It is unclear whether EFX allocations are always present in general settings, despite their conceptual appeal. A wide range of research examining EFX under structural constraints, such as specific classes of valuations or constrained interaction models, has been sparked by this fundamental open problem. As a result, a rich and rapidly developing theory has emerged around the algorithmic and existential aspects of EFX, highlighting its central importance in modern fair division.

## 1.2 Project Focus

Building on this line of work the present project investigates the existence of EFX allocations when fairness constraints are determined by a graph structure. In this model each agent is represented as a vertex and EFX envy is only required to be resolved between agents who share an edge. This means that fairness is enforced locally as well as globally making the model particularly suitable for scenarios where interactions or comparisons are limited for example social networks, hierarchical organizations or geographically separated groups.

The project specifically explores restricted classes of graphs including:

- Trees
- Bipartite graphs
- Other sparse graph families

The goal is to determine whether EFX allocations exist in these structured environments and, if so, whether they can be computed efficiently. Studying this graph-restricted version of EFX helps to understand how network constraints impact fairness guaranties and may provide new insights toward resolving the broader open problem of EFX existence.

# Chapter 2

# The Settings

We consider a set $N$ of $n$ **agents** and a set $M$ of $m$ **indivisible goods**. Each agent $i$ has an **additive valuation function** $v_i$, where $v_i(S)$ denotes the value agent $i$ assigns to a bundle $S \subseteq M$.

An **allocation** $A = (A_1, \ldots, A_n)$ assigns disjoint bundles to all agents so that all goods are assigned.

We extend this standard model by introducing an **envy graph** $G = (N, E)$. An edge $(i, j) \in E$ indicates that agents $i$ and $j$ can compare their bundles.

Our focus is on the **Graph-EFX** fairness notion: an allocation is Graph-EFX if for every edge $(i, j) \in E$, agent $i$ does not envy agent $j$ up to the removal of *any* good from $j$'s bundle.

This project investigates the **existence and computation** of Graph-EFX allocations for restricted graph classes specially **trees** and **bipartite graphs**. The survey comes here.

## 2.1 Solution Concepts

In the fair division of indivisible goods, two central fairness notions are envy-freeness and **Proportionality**. Both attempt to capture the idea that each agent should feel fairly treated but they do so in different ways.

### Envy-Freeness (EF)

**Definition 1** (Envy-Freeness). *An allocation $A$ is said to be* envy-free (EF) *if every agent values their own bundle at least as much as the bundle of any other agent. Formally,*

$$v_i(A_i) \geq v_i(A_j) \qquad \forall i, j \in N.$$

**Example.** Suppose that there are two agents and two goods. Their valuations are:

| | $g_1$ | $g_2$ |
|---|---|---|
| Agent 1 | 10 | 5 |
| Agent 2 | 6 | 9 |

Consider the allocation: $A_1 = \{g_1\}$ and $A_2 = \{g_2\}$.

**Explanation.**

- Agent 1 gets $g_1$, which they value at 10. Agent 2's item $g_2$ is worth 5 to them. So Agent 1 does **not envy** Agent 2.
- Agent 2 gets $g_2$, which they value at 9. Agent 1's item $g_1$ is worth 6 to them. So Agent 2 does **not envy** Agent 1.

Therefore, the allocation is **envy-free**. No one prefers what the other person has.

## Proportionality (PROP)

**Definition 2** (Proportionality)**.** *An allocation $A$ is* proportional (PROP) *if each agent receives a bundle worth at least their fair share of the total value of all goods. Formally,*

$$v_i(A_i) \geq \frac{v_i(M)}{n} \qquad \forall i \in N,$$

*where $n$ is the number of agents.*

**Example.** Consider three agents and three goods. Agent 1 values the goods as follows:

$$v_1(g_1) = 6, \quad v_1(g_2) = 3, \quad v_1(g_3) = 3.$$

Thus, the total value according to Agent 1 is:

$$v_1(M) = 6 + 3 + 3 = 12.$$

The proportional fair share for Agent 1 is:

$$\frac{v_1(M)}{3} = 4.$$

Suppose the allocation gives Agent 1: $A_1 = \{g_1\}$, valued at 6.

**Explanation**

- Agent 1 believes that the entire set of goods is worth 12.
- Since there are 3 people then Agent 1 expects at least one-third of that value which is 4.
- They receive item $g_1$ worth 6, which is more than their fair share.

That is why the allocation is **proportional** for Agent 1. (We would check similarly for the other agents.)

### 2.1.1 Impossibility of EF and Proportional Allocations

As we discussed earlier envy-free (EF) and proportional (PROP) allocations are not guaranteed to exist when the goods are indivisible. The simplest illustration is the case of two agents and a single indivisible item that both agents value positively: since only one agent can receive the item, the other agent obtains zero utility, envies the recipient, and also fails to receive her proportional share. (For another example where neither EF nor PROP can be accomplished we will see Example 2.)

Even if one is interested in identifying EF or PROP allocations whenever they do exist the computational difficulty remains significant. In particular, deciding whether a given instance admits an EF allocation or a proportional allocation is **NP-complete**. This hardness result follows from a standard reduction from the PARTITION problem [13] which shows that even the existence question is computationally difficult.

These fundamental impossibility and hardness results have motivated the development of several relaxations of these fairness concepts specifically tailored to the discrete setting like EF1, EFX, and approximate proportionality.

## Example: The Single Good Problem

Consider a scenario with:

- **Agents:** $N = \{1, 2\}$
- **Goods:** $M = \{g\}$ (a single indivisible good)
- **Valuations:** $v_1(g) = v_2(g) = 1$ (both agents value the good equally)

We analyze the two possible allocations:

**Case 1: Agent 1 receives the good**

- Allocation: $A_1 = \{g\}, A_2 = \emptyset$
- **Envy-Freeness:** Agent 2 envies Agent 1 since $v_2(A_2) = 0 < 1 = v_2(A_1)$
- **Proportionality:**
  - Agent 1's share: $v_1(A_1) = 1 \geq \frac{1}{2}v_1(M) = 0.5$ ✓
  - Agent 2's share: $v_2(A_2) = 0 < \frac{1}{2}v_2(M) = 0.5$ ✗

**Case 2: Agent 2 receives the good**

- Allocation: $A_1 = \emptyset, A_2 = \{g\}$
- **Envy-Freeness:** Agent 1 envies Agent 2 since $v_1(A_1) = 0 < 1 = v_1(A_2)$
- **Proportionality:**
  - Agent 1's share: $v_1(A_1) = 0 < \frac{1}{2}v_1(M) = 0.5$ ✗
  - Agent 2's share: $v_2(A_2) = 1 \geq \frac{1}{2}v_2(M) = 0.5$ ✓

In both possible allocations, at least one fairness condition is violated. This example demonstrates that, for indivisible goods, **no allocation can guaranty both envy-freeness and proportionality** in all instances. This fundamental impossibility motivates the study of relaxed fairness notions such as EF1, EFX, and MMS, which provide meaningful fairness guaranties while remaining achievable in discrete settings.

### 2.1.2 Important Concepts

Since exact envy-freeness (EF) and proportionality (PROP) may not exist for indivisible goods, several relaxations have been proposed to capture weaker but achievable notions of fairness in the discrete setting. One of the most influential relaxations is **envy-freeness up to one good (EF1)**, introduced implicitly by Lipton et al. [13] and formally defined by Budish [7].

The intuition behind EF1 is that limited envy is allowed, but only if it can be completely removed by deleting a single good from another agent's bundle. Thus, even though perfect envy-freeness may be impossible, "almost" envy-free outcomes can still be achieved.

**Definition 3** (Envy-Freeness up to One Good (EF1))**.** *An allocation $A$ is* envy-free up to one good (EF1) *if, for every pair of agents $i, j \in N$, there exists a good $g \in A_j$ such that*

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}).$$

**Example** Consider three agents and four goods with valuations shown in Table 2.1.

| | $g_1$ | $g_2$ | $g_3$ | $g_4$ |
|---|---|---|---|---|
| $a_1$ | 8 | 6 | 2 | 1 |
| $a_2$ | 5 | 4 | 7 | 3 |
| $a_3$ | 9 | 1 | 3 | 2 |

Table 2.1: Valuations of the agents for the EF1 example.

Consider the allocation:

$$A_1 = \{g_2\}, \qquad A_2 = \{g_3\}, \qquad A_3 = \{g_1, g_4\}.$$

**Explanation.**

- Agent $a_2$ does not envy anyone because she receives $g_3$, her highest-valued item.
- Agent $a_3$ also does not envy anyone since she receives $g_1$, which is best for her.
- Agent $a_1$ envies $a_3$ because $g_1$ is worth 8 to $a_1$, which is more than the value of $g_2$ (which is 6). However, if we remove $g_1$ from $A_3$, then:

  $$v_1(A_1) = 6 \quad \text{and} \quad v_1(A_3 \setminus \{g_1\}) = v_1(\{g_4\}) = 1,$$

  so the envy disappears.

Thus, every instance of envy can be eliminated by removing at most one good from the envied bundle. Therefore, this allocation satisfies **EF1**.

## A Stronger Relaxation: Envy-Freeness up to Any Good (EFX)

Although EF1 is guaranteed to exist even under very general monotone valuations, it is often viewed as a relatively weak fairness guaranty. The main limitation is that EF1 may require removing a highly valuable good from another agent's bundle to eliminate envy. In many real-world scenarios—such as dividing houses, vehicles, or other expensive resources—this may not be a meaningful or convincing notion of fairness.

For example, in the EF1 example discussed earlier, agent $a_1$ only stops envying others after the hypothetical removal of goods that she values highly. This highlights a drawback of EF1: it may permit substantial envy as long as it is resolved by deleting a single (possibly extremely valuable) item.

A more compelling relaxation, proposed to address this issue, is **envy-freeness up to any good (EFX)**. This notion was introduced by Caragiannis et al. [17] in the conference version of their work, and independently by Gourvès et al. [18] under the name *near envy-freeness*.

The requirement in EFX is stronger and more intuitive: an agent $i$ must not envy another agent $j$ after removing *any* single good from $j$'s bundle. In other words, no matter which good is removed, the envy disappears.

**Definition 4** (Envy-Freeness up to Any Good (EFX))**.** *An allocation $A$ is said to satisfy* envy-freeness up to any good (EFX) *if, for every pair of agents $i, j \in N$ and for every good $g \in A_j$, it holds that*

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}).$$

This makes EFX a much stronger fairness requirement than EF1. While EF1 guarantees the removal of *some* good eliminates envy, EFX requires that the removal of *every* good in the envied bundle eliminate envy. Unfortunately, unlike EF1, it remains an open question whether EFX allocations always exist, even for simple additive valuations.

**Example** Consider an instance with three agents $a_1, a_2, a_3$ and five goods $g_1, g_2, g_3, g_4, g_5$. Suppose the agents have the following (strict) rankings:

$$\begin{aligned} a_1 :\ & g_1 \succ g_2 \succ g_3 \succ g_4 \succ g_5, \\ a_2 :\ & g_3 \succ g_1 \succ g_5 \succ g_4 \succ g_2, \\ a_3 :\ & g_2 \succ g_4 \succ g_5 \succ g_3 \succ g_1. \end{aligned}$$

Consider the allocation

$$A_1 = \{g_2, g_5\}, \qquad A_2 = \{g_1, g_4\}, \qquad A_3 = \{g_3\}.$$

This allocation is *not* EFX. In particular, agent $a_3$ envies $a_2$, and this envy cannot be eliminated by removing $g_4$, which is $a_3$'s least valued good in $A_2$.

However, it is easy to obtain an EFX allocation by slightly modifying the bundles:

$$B_1 = \{g_1, g_5\}, \qquad B_2 = \{g_3, g_4\}, \qquad B_3 = \{g_2\}.$$

In this allocation, the envy of $a_1$ toward $a_3$ can be eliminated by removing $g_2$ from $B_3$, while the envy of $a_2$ toward $a_1$ can be eliminated by removing $g_5$ from $B_1$. In both cases, the removed item is the least valued good (from the perspective of the envious agent) in the other agent's bundle. Therefore, $B$ is an EFX allocation.

## EFX and Maximin Share Fairness

The existence of EFX allocations continues to be one of the most challenging open problems in fair division. Procaccia [12] famously referred to this as "fair division's most enigmatic question," highlighting the difficulty of establishing whether an EFX allocation exists for all instances. Although recent research has provided partial results and approximate guarantees for special classes of problems, a full resolution remains open (see Section 4).

Beyond EF1 and EFX, another extensively studied concept in the allocation of indivisible goods is the *maximin share* (MMS), introduced by Budish [7]. This notion generalizes the intuition behind the classical "cut-and-choose" procedure used for divisible resources.

**Definition 5** (Maximin Share (MMS) [7])**.** *For an agent $i$ with valuation function $v_i$ and a set of goods $M$, the* ***maximin share*** *of agent $i$ is defined as*

$$\mu_i^n(M) = \max_{(B_1,\dots,B_n)} \min_{k\in\{1,\dots,n\}} v_i(B_k),$$

*where the maximum is taken over all possible partitions of $M$ into $n$ bundles.*

Intuitively, the maximin share represents the value an agent can guarantee herself if she were to divide the goods into $n$ bundles and then receive the *worst* bundle from her own partition. Thus it is considered a significant relaxation of proportionality to guarantee that every agent receives a bundle worth at least their MMS value particularly in situations where items cannot be divided.

**Example.** Consider three agents $a_1, a_2, a_3$ and four goods $g_1, g_2, g_3, g_4$ with additive valuations given in Table 2.2.

| | $g_1$ | $g_2$ | $g_3$ | $g_4$ |
|---|---|---|---|---|
| $a_1$ | 6 | 3 | 2 | 1 |
| $a_2$ | 4 | 4 | 3 | 2 |
| $a_3$ | 5 | 2 | 2 | 1 |

Table 2.2: Valuations for the MMS example.

We compute the maximin share for agent $a_1$ (the same procedure applies to others). Agent $a_1$'s total value is $v_1(M) = 6+3+2+1 = 12$. For $n = 3$ agents, agent $a_1$ considers all partitions of the four goods into three bundles and selects the partition that maximizes the value of the worst bundle (from her perspective). One candidate partition is:

$$B_1 = \{g_1\}, \quad B_2 = \{g_2\}, \quad B_3 = \{g_3, g_4\}.$$

Under this partition, the bundle values for $a_1$ are $v_1(B_1) = 6$, $v_1(B_2) = 3$, $v_1(B_3) = 3$; the worst bundle is worth 3. It is easy to check that no partition gives $a_1$ a worst bundle worth more than 3, so $\mu_1^3(M) = 3$.

Now consider the allocation

$$A_1 = \{g_1\}, \quad A_2 = \{g_2\}, \quad A_3 = \{g_3, g_4\}.$$

Agent $a_1$ receives value $v_1(A_1) = 6 \geq \mu_1^3(M) = 3$, so the MMS condition holds for $a_1$. Similarly, computing $\mu_2^3(M)$ and $\mu_3^3(M)$ (by checking partitions from each agent's viewpoint) shows that the same allocation gives every agent at least her maximin share in this instance; hence it is an MMS allocation.

Intuitively, each agent imagine themselves cutting the goods into three piles and then being forced to take the least attractive pile. The maximin share is the best guarantee that they can secure with this pessimistic strategy. In the example above $a_1$ we can ensure at least value 3 by splitting as $\{g_1\}, \{g_2\}, \{g_3, g_4\}$ — the worst pile is value 3, so any actual allocation that gives $a_1$ value 3 or more meets her MMS. The allocation chosen here gives $a_1$ value 6, comfortably above her MMS.

## Computational Complexity and Existence of MMS Allocations

Although the maximin share is conceptually simple computing an agent's MMS value is computationally challenging. In fact, determining the exact maximin share or finding an allocation that satisfies all agents MMS guarantees is NP-hard problem as shown through a straightforward reduction from the PARTITION problem. However Woeginger [4] established that there exists a polynomial-time approximation scheme (PTAS) for computing the MMS value of an agent.

Since this method was introduced a major question in fair division was whether an MMS allocation always exists. This remained open for several years until the breakthrough results of Kurokawa, Procaccia, and Wang [5, 6], who demonstrated that MMS allocations do *not* always exist when there are three or more agents. Despite this impossibility, subsequent work has focused on computing *approximate* MMS allocations, which guarantee each agent a fraction of her MMS value and are often achievable efficiently.

# Chapter 3

# Envy-Freeness up to One Goods (EF1)

## 3.1 Simple Algorithms for EF1 Allocation

A number of simple and efficient procedures are known to compute envy-free up to one good (EF1) allocations. One of the most well-known among them is the *Round-Robin* algorithm which operates in a sequence of rounds. At the beginning of the process the agents are arranged in a fixed arbitrary order. During each round following this order every agent selects her most preferred good among those that remain unallocated. This continues until all goods have been assigned.

The appeal of the Round-Robin procedure lies in both its simplicity and its fairness guarantee. As shown by Caragiannis et al. [2], the allocation produced by Round-Robin is always EF1. In other words although agents may envy others, such envy can always be eliminated by removing at most one item from the envied agent's bundle.

**Algorithm 1** Round-Robin

**Input:** A fair allocation instance $I = \langle N, M, v \rangle$ with $n$ agents and $m$ goods.
**Output:** Allocation $A = (A_1, \ldots, A_n)$.
**for** each agent $i \in N$ **do**
  $A_i \leftarrow \varnothing$;
**end**
**for** $t = 1, \ldots, m$ **do**
  Let $i \leftarrow t \bmod n$;
  Let $g^* \in \arg\max_{g \in M} v_i(g)$;
  $A_i \leftarrow A_i \cup \{g^*\}$;
  $M \leftarrow M \setminus \{g^*\}$;
**end**

**Theorem 1** (Round-Robin is EF1; Caragiannis et al. [2]). *Round-Robin (agents pick in a fixed order each round each agent chooses her most-preferred remaining good) always returns an EF1 allocation.*

*Proof.* We prove by induction on $m = |M|$, the number of goods.

**Base.** If $m = 0$ or $m = 1$ the claim is immediate: with no goods the allocation is trivially EF (hence EF1); with one good the agent who receives it may be envied by others but removing that single good from the recipient's bundle eliminates any envy so EF1 holds.

**Inductive step.** Assume the statement holds for all instances with fewer than $m$ goods, and consider an instance with $m \geq 2$ goods and an arbitrary fixed ordering of the $n$ agents. Run Round-Robin for one full pass (the first round). Let $a$ denote the first agent in the fixed order and let $g$ be the first good she picks. Remove $g$ from the set of goods and consider the execution of Round-Robin on the remaining set $M' = M \setminus \{g\}$.

Observe that after the removal of $g$ the residual procedure can be considered as a Round-Robin execution on $M'$ with the first picker being the agent who follows $a$ in the initial order, since $a$ has already selected her item in the first round. In particular, for any two agents $(i, j)$ the order in which they will select in the run on $M'$ is either identical to that in the original instance or is cyclically shifted; crucially, the run on $M'$ respects the same Round-Robin rule, and hence by the induction hypothesis it returns an allocation $A'$ of $M'$ which is EF1 w.r.t. the picks on $M'$.

Let $A$ be the final allocation of the full Round-Robin run on $M$; $A$ equals $A'$ except that $a$'s bundle in $A$ is $A_a = A'_a \cup \{g\}$ while all other bundles coincide with those in $A'$.

We now show $A$ is EF1. Take any ordered pair of agents $(i, j)$.

- If $i = a$ (the agent who picked $g$ first) then it is $i$ picked before $j$ in every round of the algorithm. By the standard Round-Robin logic (and because $i$ always chose her most preferred available item when it was her turn), $i$ cannot strictly prefer $j$'s bundle to their own: whenever $j$ chose an item, $i$ had an earlier opportunity in that same round to choose an item she preferred at least as much. Thus $v_i(A_i) \geq v_i(A_j)$ and so $i$ does not envy $j$ (in particular $i$ satisfies EF1 trivially).

- If $i \neq a$, consider two subcases.

  1. If $g \notin A_j$, then $A_j = A'_j$ and $A_i = A'_i$. Since $A'$ is EF1 by the induction hypothesis, there exists some $h \in A'_j$ (possibly none if no envy) such that

  $$v_i(A'_i) \geq v_i(A'_j \setminus \{h\}).$$

     Hence the same good $h$ serves to witness EF1 for $i$ vs. $j$ in $A$ (because the bundles are identical to those in $A'$).

  2. If $g \in A_j$, then $j = a$ (only $a$ received $g$), contradicting $i \neq a$. Therefore this case does not occur.

The only remaining potential source of envy introduced by adding $g$ to $a$'s bundle is envy of some agent $i \neq a$ towards $a$. But by the argument above the allocation on $M'$ was EF1 for the pair $(i, a)$, so there exists $h \in A'_a$ (possibly none) with

$$v_i(A'_i) \geq v_i(A'_a \setminus \{h\}).$$

If this $h$ exists, it also witnesses EF1 for $(i, a)$ in $A$ because $A_a = A'_a \cup \{g\}$ and thus removing $h$ (which belongs to $A'_a$) from $A_a$ leaves at most the additional item $g$; even if $g$ makes $A_a$ more attractive to $i$, we can choose as witness the first item $g$ that $a$ picked (namely the initial $g$): removing that particular $g$ from $A_a$ leaves $A'_a$, and by the EF1 property of $A'$ we have

$$v_i(A_i) = v_i(A'_i) \geq v_i(A'_a \setminus \{h\}) = v_i((A_a \setminus \{g\}) \setminus \{h\}) \geq v_i(A_a \setminus \{g\}),$$

so removing the single good $g$ from $A_a$ eliminates $i$'s envy. Thus every envy of some agent $i$ toward $a$ can be eliminated by removing at most one good (namely $g$) from $a$'s bundle.

Combining the cases above, for every ordered pair $(i, j)$ there exists some single good in $A_j$ whose removal eliminates $i$'s envy toward $j$. Therefore $A$ is EF1, completing the induction. $\square$

## Generalizing Round-Robin Sequences

In the standard version of the Round–Robin procedure, the agents appear in the *same* fixed order in every round and each agent selects her most-preferred remaining good when it is their turn. But this fixed ordering is not essential to guaranty the fairness of EF1. As observed by Caragiannis et al. [2], the procedure still produces an EF1 allocation even when the order of agents is allowed to change from round to round provided that each agent always chooses their favorite available good when the turn comes.

This flexibility leads to a broader family of EF1-guaranteeing algorithms based on *recursively balanced sequences*. In such sequences the number of turns taken by any two agents never differs by more than one at any time during the process. Intuitively, this ensures that no agent can "fall too far behind" in the selection process, thereby preserving the structural property that enables EF1 fairness. Consequently, Round–Robin can be viewed simply as one particular instance of this larger class of balanced-turn algorithms that all compute EF1 allocations under the same fundamental principle of balanced opportunity.

### 3.1.1 Envy-Cycle Elimination Algorithm

A second fundamental approach for computing EF1 allocations is the *Envy–Cycle Elimination* algorithm, originally introduced by Lipton et al. [22]. Unlike Round-Robin or other sequential picking procedures, this method does not rely on a predetermined ordering of agents. Instead, the algorithm repeatedly identifies agents who are currently at a disadvantage and assigns them their most-preferred unallocated good.

The algorithm maintains an *envy graph*, where each vertex represents an agent, and a directed edge $i \to j$ indicates that agent $i$ strictly prefers agent $j$'s bundle over her own. Every time the algorithm runs, it tries to assign an unassigned good to an agent whose vertex has in-degree zero or an agent that no one envies. Such an agent exists whenever the envy graph is acyclic.

If no such agent exists, then the envy graph necessarily contains a directed cycle. Let

$$C = (i_1, i_2, \ldots, i_d)$$

be a directed cycle, meaning that each $i_t$ envies $i_{t+1}$ for $t = 1, \ldots, d-1$, and $i_d$ envies $i_1$. The cycle can be eliminated by performing a cyclic update of the bundles: every agent in the cycle receives the bundle of the agent she points to. Formally,

$$A_{i_t} \leftarrow A_{i_{t+1}} \qquad \text{for } t = 1, \ldots, d-1, \quad \text{and} \quad A_{i_d} \leftarrow A_{i_1}.$$

This rotation strictly reduces total envy on the cycle and breaks the cycle in the envy graph.

By repeatedly allocating goods to non-envied agents and eliminating envy cycles through such rotations, the algorithm continues until all goods have been assigned. Lipton et al. [13] proved that this procedure always terminates and always returns an EF1 allocation.

### Envy Graph Illustration

Below we show a simple depiction of an envy graph containing a directed cycle, which would trigger a cycle-elimination step in the algorithm.


**Algorithm 2** Envy-Cycle Elimination

**Input:** A fair allocation instance $I = \langle N, M, v \rangle$ with $n$ agents and $m$ goods.
**Output:** Allocation $A = (A_1, \ldots, A_n)$.
**for** each agent $i \in N$ **do**
  $A_i \leftarrow \varnothing$;
**end**
**for** $t = 1, \ldots, m$ **do**
  **while** there does not exist an unenvied agent **do**
    Find an envy-cycle $C = (i_1, \ldots, i_d)$ and resolve the cycle as follows:

$$A_i^C = \begin{cases} A_{i_{j+1}} & \text{for all } 1 \leq j \leq d-1 \\ A_{i_1} & \text{for } j = d \end{cases} \tag{1}$$

    $A_i \leftarrow A_i^C$ for all $i \in C$;
  **end**
  Let $i$ be an unenvied agent;
  Let $g^* \in \arg\max_{g \in M} v_i(g)$;
  $A_i \leftarrow A_i \cup \{g^*\}$;
  $M \leftarrow M \setminus \{g^*\}$;
**end**

## Correctness of the Envy-Cycle Elimination Algorithm

In this section we establish the correctness of the Envy-Cycle Elimination algorithm introduced by Lipton et al. [13]. Specifically, we show that the algorithm (1) terminates in polynomial time and (2) always outputs an EF1 allocation.

**Theorem 2.** *The Envy-Cycle Elimination algorithm runs in polynomial time and produces an allocation satisfying envy-freeness up to one good (EF1).*

*Proof.* We divide the proof into three main parts: termination, complexity, and EF1 correctness.

**1. Termination.** The algorithm allocates exactly one good in each iteration. Since no good is ever reassigned except during cycle elimination, the total number of allocations is at most $m$, where $m$ is the number of goods. Cycle elimination steps only reassign goods among agents but do not introduce new unallocated goods. Additionally every cycle-elimination strictly reduces the number of directed cycles in the envy graph meaning that a specific cycle can never reappear in the exact same form. Thus, the algorithm performs a finite number of cycle removals, ensuring that it must terminate after at most $m$ allocation steps.

**2. Polynomial-Time Complexity.** Each iteration of the algorithm requires:

- identifying an agent of in-degree zero, or detecting a cycle in the envy graph,
- assigning a single good,
- and possibly performing one cycle-elimination operation.

Both cycle detection and cycle elimination on a graph of $n$ nodes can be done in polynomial time. Because the algorithm performs at most $m$ iterations and a polynomial number of cycle eliminations the total running time remains polynomial in $n + m$, as shown in [22].

**3. EF1 Correctness.** To prove that the final allocation satisfies EF1, we focus on how envy is controlled throughout the execution.

At any step a good is always assigned to an agent who is *not* currently envied by any other agent—that is, a vertex of in-degree zero in the envy graph. This ensures the following fundamental property:

> *When an agent receives a new good, removing that good eliminates all envy towards her.*

Why is this true? Because before the allocation they had in-degree zero and hence no one envied them. Thus the only possible cause of envy would be the newly assigned good, so removing it restores the envy-free status.

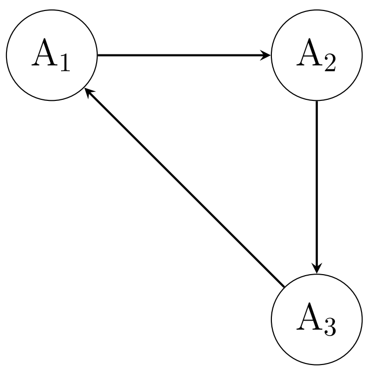


Figure 3.1: An envy graph containing a directed cycle ($A_1 \to A_2 \to A_3 \to A_1$). Cycle elimination redistributes bundles along the cycle to break it.

Now consider the second case: when the algorithm encounters a cycle in the envy graph. Cycle elimination rotates bundles along the directed cycle. For a cycle $(i_1, i_2, \ldots, i_d)$, each agent $i_k$ receives the bundle of the agent she envies $(i_{k+1})$. Because every edge of the cycle represents envy, each agent strictly prefers the bundle she receives over her old one. Thus:

- Every agent in the cycle is strictly better off.
- No new envy is introduced along the cycle.
- The cycle is eliminated permanently.

This results in an updated envy graph that is acyclic, enabling the algorithm to resume picking an agent of in-degree zero.

Finally, once all goods are allocated, every agent either:

- never receives a good that causes persistent envy, or
- receives goods whose last addition can be removed to eliminate all possible envy.

Therefore, the allocation satisfies the EF1 condition. □

□

## 3.2 Pareto Optimal (PO) Allocations & Maximum Nash Welfare (MNW)

Apart from ensuring fairness, it is natural to demand that an allocation be *efficient*. Caragiannis et al. [2] identified a compelling connection between EF1 allocations, Pareto optimality (PO), and the celebrated maximum Nash welfare (MNW) rule. In this subsection, we revisit these notions with formal definitions and an illustrative diagram.

**Definition 6** (Pareto Optimality (PO)). *An allocation $A = (A_1, \ldots, A_n)$ is said to be* Pareto optimal *if there exists no other allocation $B$ such that:*

$$v_i(B_i) \geq v_i(A_i) \quad \text{for all } i \in N,$$

*and*

$$v_j(B_j) > v_j(A_j) \quad \text{for some } j \in N.$$

*Equivalently, $A$ is Pareto optimal if it is not Pareto dominated by any other allocation.*

**Definition 7** (Maximum Nash Welfare (MNW) Allocation). *An allocation $A$ is a* maximum Nash welfare (MNW) allocation *if it simultaneously satisfies:*

1. *it maximizes the number of agents receiving strictly positive total value;*
2. *among all such allocations, it maximizes the Nash social welfare:*

$$NSW(A) \;=\; \prod_{i:v_i(A_i)>0} v_i(A_i).$$

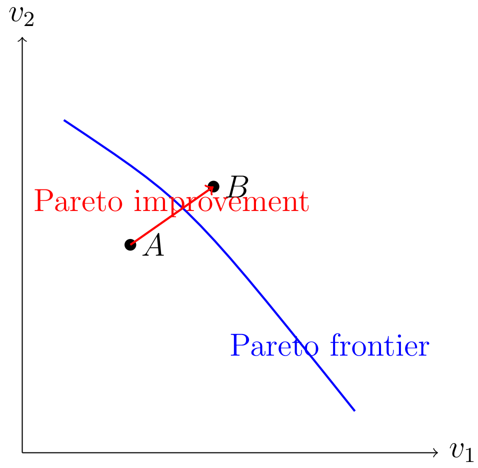


Figure 3.2: A geometric view of Pareto improvements. A point $B$ dominates $A$ if all agents weakly gain and at least one gains strictly. The blue curve represents the Pareto frontier.

MNW allocations have been shown to satisfy both EF1 and Pareto optimality for additive valuations, establishing a strong link between fairness and efficiency. Maximising the geometric mean of utilities intuitively ensures that no resources are wasted while preventing allocations that are excessively unfair.

The figure illustrates the idea of Pareto improvements: moving from allocation $A$ to $B$ increases utility for at least one agent while harming none making $A$ Pareto dominated. Allocations on the Pareto frontier cannot be improved without hurting someone, thereby satisfying the efficiency criterion.

## Theorem 2 (Caragiannis et al. [2])

Every MNW allocation is EF1 and Pareto optimal (PO).

*Proof.* We prove the two properties separately.

**Pareto optimality.** Let $A$ be an MNW allocation and suppose, toward a contradiction, that $A$ is not Pareto optimal. Then there exists an allocation $B$ such that $v_i(B_i) \geq v_i(A_i)$ for all $i \in N$ and $v_j(B_j) > v_j(A_j)$ for some $j \in N$. Consider the Nash welfare (product of positive utilities) of $B$ and $A$. Because every agent's utility in $B$ is at least her utility in $A$, and at least one agent's utility is strictly larger, it follows that the Nash welfare of $B$ is strictly larger than that of $A$. This contradicts the definition of $A$ as an allocation that maximizes Nash welfare. Hence $A$ must be Pareto optimal.

**EF1 correctness.** Assume, for contradiction, that an MNW allocation $A$ is not EF1. Then there exist two agents $i$ and $j$ such that for every good $g \in A_j$,

$$v_i(A_i) < v_i(A_j \setminus \{g\}).$$

Equivalently, for every $g \in A_j$,

$$v_i(A_i) + v_i(g) < v_i(A_j). \tag{1}$$

We will construct a new allocation $A'$ obtained from $A$ by moving a single item $g^* \in A_j$ from agent $j$ to agent $i$, and we will show that this move strictly increases the Nash welfare, contradicting the optimality of $A$.

Choose an item $g^* \in A_j$ that minimizes the ratio

$$\frac{v_j(g)}{v_i(g)} \qquad \text{over } g \in A_j$$

(with the convention that if some $v_i(g) = 0$ we treat the ratio as $+\infty$ and avoid such items unless all have $v_i(g) = 0$ — the argument below handles the zero-valued cases separately). Intuitively, $g^*$ is the item in $A_j$ that is "cheapest" for $j$ relative to its value for $i$.

Define allocation $A'$ by

$$A'_i = A_i \cup \{g^*\}, \qquad A'_j = A_j \setminus \{g^*\},$$

and $A'_\ell = A_\ell$ for all $\ell \notin \{i, j\}$.

We compare the contribution of agents $i$ and $j$ to the Nash welfare before and after the transfer. Let

$$x := v_i(A_i), \qquad y := v_j(A_j), \qquad \alpha := v_i(g^*), \qquad \beta := v_j(g^*).$$

The product of utilities for $i$ and $j$ in $A$ is $x \cdot y$, while in $A'$ it becomes $(x + \alpha) \cdot (y - \beta)$. We will show

$$(x + \alpha)(y - \beta) > xy,$$

which is equivalent to

$$\frac{\alpha}{x} > \frac{\beta}{y}. \tag{2}$$

Thus it suffices to prove (2). By the choice of $g^*$ as minimizing $\beta/\alpha$ over items in $A_j$, we have for every $g \in A_j$

$$\frac{v_j(g)}{v_i(g)} \ \geq \ \frac{\beta}{\alpha}.$$

Summing the inequality $v_j(g) \geq \frac{\beta}{\alpha} v_i(g)$ over all $g \in A_j$ gives

$$y = \sum_{g \in A_j} v_j(g) \ \geq \ \frac{\beta}{\alpha} \sum_{g \in A_j} v_i(g) \ = \ \frac{\beta}{\alpha} v_i(A_j).$$

Hence

$$\frac{\beta}{y} \leq \frac{\alpha}{v_i(A_j)}. \tag{3}$$

Using assumption (1) (which holds for $g^*$ in particular) we have

$$x + \alpha < v_i(A_j).$$

Therefore

$$\frac{\alpha}{x} \ > \ \frac{\alpha}{v_i(A_j)} \ \geq \ \frac{\beta}{y},$$

where the last inequality follows from (3). This proves (2), and hence

$$(x + \alpha)(y - \beta) > xy.$$

Consequently, the two-agent product increases while the utilities of other agents are unchanged; therefore the overall Nash welfare strictly increases from $A$ to $A'$. This contradicts the maximality of Nash welfare for $A$. Thus our assumption that $A$ is not EF1 is false, and $A$ must be EF1.

Combining the two parts above, every MNW allocation is both Pareto optimal and EF1. $\square$

**Example 1** (A Single Example Illustrating Both PO and MNW)**.** *Consider two agents $a_1$ and $a_2$ and three goods $g_1, g_2, g_3$. Their additive valuations are given in Table 3.1.*

| **Agent** | $g_1$ | $g_2$ | $g_3$ |
|---|---|---|---|
| $a_1$ | 6 | 4 | 1 |
| $a_2$ | 3 | 5 | 2 |

Table 3.1: Valuations of the agents.

*Consider the following allocation:*

$$A_1 = \{g_1\}, \qquad A_2 = \{g_2, g_3\}.$$

**Utilities.**

$$v_1(A_1) = 6, \qquad v_2(A_2) = 5 + 2 = 7.$$

*The Nash welfare of this allocation is:*

$$NSW(A) = v_1(A_1) \cdot v_2(A_2) = 6 \times 7 = 42.$$

**Why this allocation is MNW.** *Let us check that no other allocation gives a higher Nash welfare.*

- *If both $g_1$ and $g_2$ go to $a_1$, then $a_2$ receives only $g_3$:*

$$(v_1, v_2) = (10, 2), \quad NSW = 20.$$

- *If both $g_2$ and $g_3$ go to $a_1$, then $a_2$ receives only $g_1$:*

$$(v_1, v_2) = (5, 3), \quad NSW = 15.$$

- *If $g_1$ goes to $a_2$ and the others to $a_1$:*

$$(v_1, v_2) = (5, 3), \quad NSW = 15.$$

*None of these welfare products exceed* 42*. Hence, the allocation above* maximizes the Nash welfare *and is therefore an* MNW allocation.

**Why the allocation is also Pareto optimal (PO).** *Suppose there exists another allocation $B$ such that:*

$$v_1(B_1) \geq 6, \qquad v_2(B_2) \geq 7,$$

*with at least one strict inequality. However:*

- *Agent* 1 *receives their top good* $g_1$ *(value 6); hence it cannot reach value* $> 6$ *because the total value of all other goods to them is only* $4 + 1 = 5$.
- *Agent* 2 *receives goods giving her value* 7*, which is the maximum they can get because* $5 + 2 = 7$.

*Thus, no alternative allocation can make one agent strictly better off without harming the other. Therefore, the MNW allocation is also* Pareto optimal.

## Fairness and Efficiency: The Role of Nash Welfare

The discussion so far highlights a remarkable fact in fair division: although fairness requirements such as EF1 do not always align with efficiency criteria, there exist allocations that simultaneously satisfy both properties. In particular, the work of Caragiannis et al. [2] demonstrates that allocations maximizing the Nash welfare achieve both EF1 and Pareto optimality (PO), thereby striking a strong balance between fairness and social efficiency.

A recent contribution by Yuen and Suksompong [25] further strengthens this connection by proving that the *Nash welfare* is the *unique* welfarist objective whose maximization guarantees EF1 and PO allocations. This establishes Nash welfare as a mathematically distinguished and principled notion for fair allocation among agents with additive valuations.

However, despite its elegant fairness–efficiency guarantees, computing a maximum Nash welfare (MNW) allocation is computationally challenging. It is known that approximating MNW in polynomial time is hard in general (see the discussion in Section 7.1). Barman et al. [8] made partial progress by designing a pseudo-polynomial-time algorithm, yet it remains an important open question whether MNW allocations can be computed in fully polynomial time.

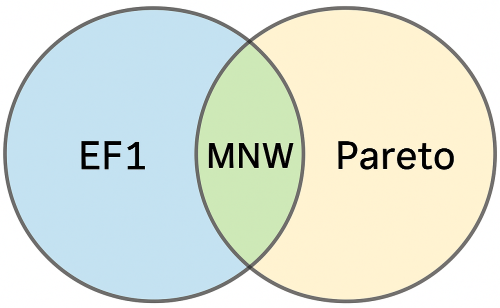


Figure 3.3: A conceptual illustration of fairness–efficiency interactions: MNW allocations lie at the intersection of EF1 and Pareto optimality.

# Chapter 4

# Envy Freeness up to Any Goods (EFX)

In contrast to the case of EF1 allocations—whose existence is guaranteed by simple polynomial-time procedures—the question of whether EFX allocations always exist remains one of the most difficult open problems in fair division. Over the past few years, several works have established the existence of EFX allocations for a variety of important restricted settings. These results typically fall into two main categories: scenarios involving a small number of agents and instances in which the valuation functions satisfy particular structural assumptions. In this section, we summarize the known existence guaranties for these special cases and subsequently examine recent progress on approximate variants of EFX, which relax the fairness requirement while preserving strong fairness guaranties.

## 4.1 Established Existence Results for EFX in Restricted Domains

We begin by examining several important scenarios in which the existence of EFX allocations has been established. These special cases provide valuable insight into when the strong fairness guarantee of EFX can be achieved and form the basis for broader investigation into the general problem.

### 4.1.1 Identical Valuations and the Leximin++ Solution

When all agents share the same valuation functionthat is, $v_i(\cdot) = v(\cdot)$ for every agent $i$the existence of EFX allocations is significantly easier to guarantee. Plaut and Roughgarden [3] established one of the first positive results in this setting by introducing a refined variant of the classical leximin solution known as the *leximin++* allocation.

**Leximin and Leximin++.** A standard leximin allocation maximizes the minimum bundle value, then the second minimum bundle value and so on. But such an allocation is *not* always EFX. To address this, Plaut and Roughgarden introduced the leximin++ rule, which adds an additional tie-breaking requirement:

> **Leximin++ refinement:** After maximizing the minimum value, the allocation must also maximize the size of the minimum-value bundle before

proceeding to maximize the second minimum value.

This refinement is crucial for enforcing EFX.

**Key Properties of Leximin++ Allocations.** A leximin++ allocation $A$ under identical valuations satisfies:

1. **Strong fairness:** It optimizes the welfare of the worst-off agent and then refines the order of agents by bundle size.
2. **Monotonicity:** Increasing the value of a good cannot harm any agent in the leximin++ ordering.
3. **EFX Guarantee:** Every leximin++ allocation is EFX, even when $v(\cdot)$ is non-additive.

**Correctness: Why Leximin++ Is EFX.** Let the bundles be ordered so that

$$v(A_1) \leq v(A_2) \leq \cdots \leq v(A_n).$$

Assume for contradiction that $A$ is *not* EFX. Then there exist agents $i < j$ and some $g \in A_j$ such that

$$v(A_i) < v(A_j \setminus \{g\}) \qquad \text{for every } g \in A_j.$$

Two cases arise:

1. If $v(g) > 0$, then removing $g$ from $A_j$ improves the $i$-th minimum value, contradicting leximin++ optimality.
2. If $v(g) = 0$, then transferring $g$ from $A_j$ to $A_i$ increases the size of the minimum bundle, again contradicting leximin++ tie-breaking.

In both cases, we obtain a strictly better leximin++ ordering than $A$, which is impossible. Thus, $A$ must satisfy EFX.

□

**Example.** Consider three goods $\{a, b, c\}$ and two agents with identical valuations:

$$v(a) = 4, \quad v(b) = 3, \quad v(c) = 1.$$

**Leximin allocation:**

$$A_1 = \{c\}, \quad A_2 = \{a, b\}.$$

But this is *not* EFX, because removing any good from $A_2$ leaves:

$$v(A_2 \setminus \{a\}) = 3 > 1, \qquad v(A_2 \setminus \{b\}) = 4 > 1.$$

**Leximin++ allocation:**

$$A_1 = \{b, c\}, \qquad A_2 = \{a\}.$$

Check EFX:

$$v(A_1) = 4, \quad v(A_2) = 4.$$

Removing any good from either bundle does not cause envy, so the allocation is EFX.

This example highlights how leximin++ improves upon standard leximin to achieve EFX under identical valuations.

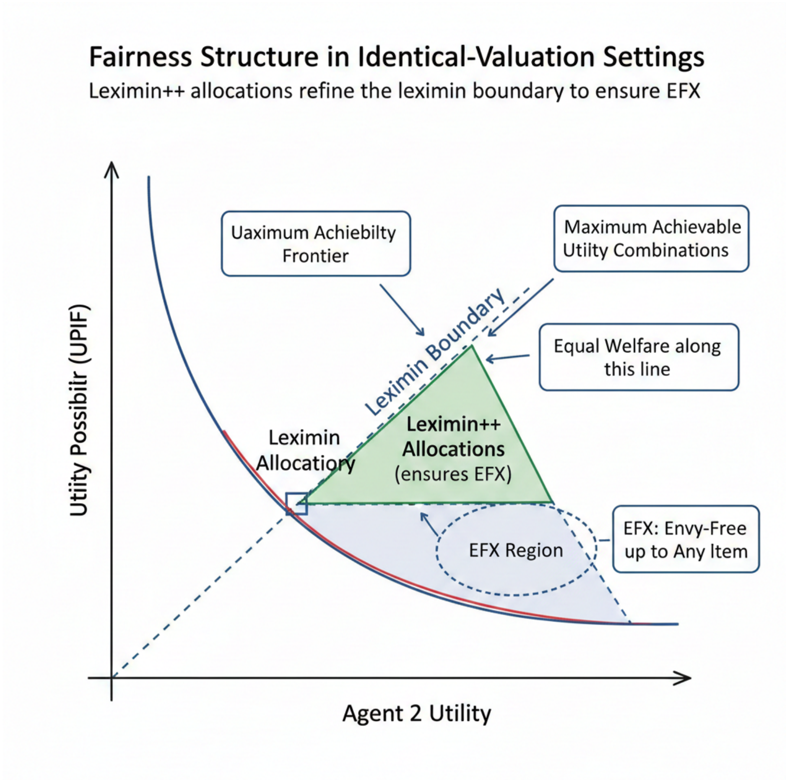


Figure 4.1: Illustration of the fairness structure in identical-valuation settings. Leximin++ allocations refine the leximin boundary to ensure EFX.

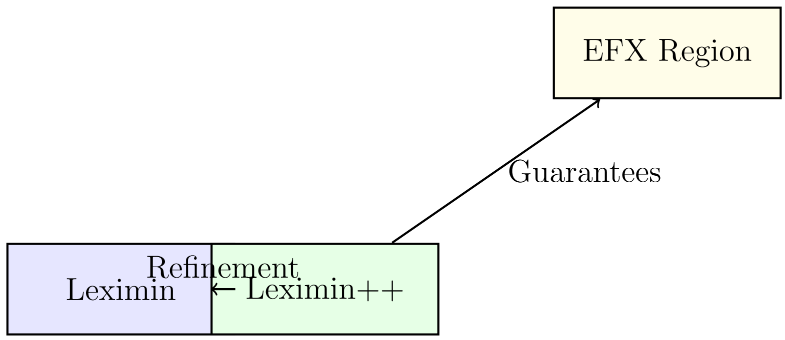


Figure 4.2: Leximin++ improves the baseline leximin rule to guarantee EFX under identical valuations.

### 4.1.2 Ordered Valuations and EFX via Envy-Cycle Elimination

Beyond the case of identical valuations, Plaut and Roughgarden [3] proved that the well-known *Envy-Cycle Elimination (ECE)* algorithm also produces an EFX allocation for a broader class of instances known as *ordered valuations*.

**Ordered Valuations.** An instance is said to have ordered valuations if all agents agree on the *relative ranking* of the goods, even though their numerical valuations may differ. Formally, the goods can be indexed as

$$g_1, g_2, \dots, g_m$$

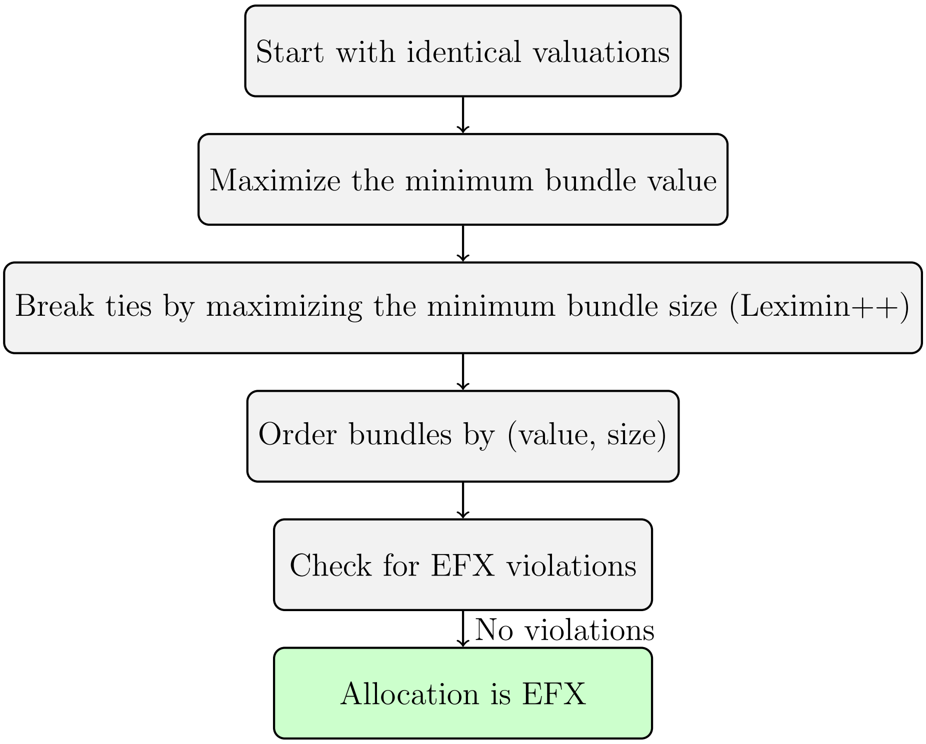


Figure 4.3: Flowchart of the Leximin++ algorithm. The refinement step ensures EFX.

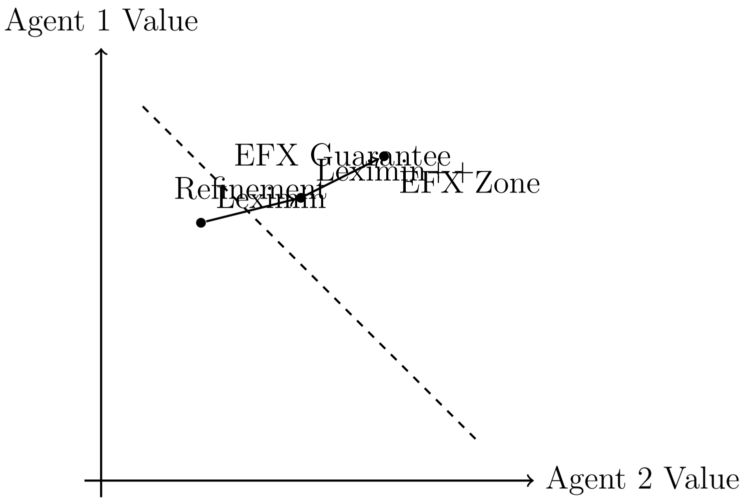


Figure 4.4: Vector-style geometric illustration of how Leximin++ moves the allocation towards the EFX-guaranteed region.

such that for every agent $i$,

$$v_i(g_1) \geq v_i(g_2) \geq \cdots \geq v_i(g_m).$$

Thus, all agents value $g_1$ the most, $g_m$ the least, although the exact values may differ.

**Why ECE Works in This Setting.** The ECE algorithm repeatedly:

1. lets a non-envied agent pick a good,

2. assigns this good to the agent,

3. eliminates any envy cycles that arise.

Because the selected good at every step is the *least* remaining good for every agent, removing it from the chosen agent's bundle is always sufficient to eliminate any envy toward that agent. This ensures that the allocation produced is EFX.

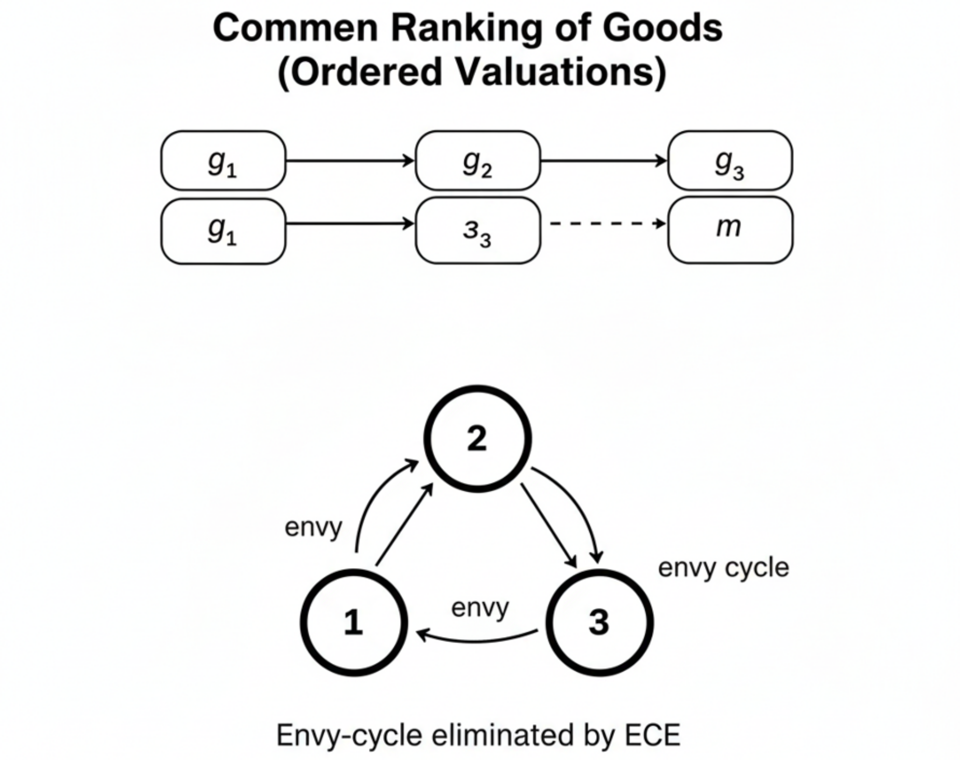


Figure 4.5: Ordered valuations: all agents have the same ranking of goods. The Envy-Cycle Elimination algorithm assigns goods from least to most valuable, ensuring the result is EFX.

**Example.** Consider three goods with a common ranking:

$$g_1 \succ g_2 \succ g_3$$

and two agents with the following valuations:

$$v_1(g_1) = 9, \quad v_1(g_2) = 6, \quad v_1(g_3) = 1,$$
$$v_2(g_1) = 15, \quad v_2(g_2) = 10, \quad v_2(g_3) = 3.$$

**ECE allocation process:**

1. Initially no one is envied. Agent 1 picks the least remaining good $g_3$.

$$A_1 = \{g_3\}, \quad A_2 = \emptyset$$

2. Agent 2 is not envied, so she picks $g_2$.

$$A_1 = \{g_3\}, \quad A_2 = \{g_2\}$$

3. The remaining good $g_1$ goes to the agent who is currently not envied (which is still agent 2).

$$A_1 = \{g_3\}, \quad A_2 = \{g_2, g_1\}$$

**EFX check:**

For agent 1:

$$v_1(g_2) = 6 > v_1(g_3) = 1, \quad v_1(g_1) = 9 > 1,$$

but

$$v_1(\{g_2, g_1\} \setminus \{g\}) \geq v_1(g_3).$$

Similarly, removing any good from agent 2's bundle removes all envy.

Thus, the allocation is EFX.

This example demonstrates how ordered valuations guaranty that ECE produces EFX allocations.

**Theorem 3** (Plaut and Roughgarden, 2020)**.** *For every ordered instance the Envy-Cycle Elimination Algorithm (Algorithm 2) computes an allocation that satisfies* EFX.

*Proof.* Let $A$ be the final allocation produced by the algorithm. Fix any pair of agents $i$ and $j$, and let $g$ be the *last* good assigned to agent $j$ during the execution.

**Step 1: Non-envy at the moment $g$ is assigned.** When good $g$ is given to agent $j$, the algorithm selects $j$ only if she is *not envied* by any other agent. Thus, for every agent $i$, we had at that moment:

$$v_i(A_i) \;\geq\; v_i(A_j \setminus \{g\}).$$

**Step 2: Ordered valuations preserve relative value of earlier goods.** Because the instance is ordered, all agents agree on which goods are "larger" or "smaller." In particular, if $g'$ is any good allocated to $j$ *before* $g$, then:

$$v_i(g') \;\geq\; v_i(g), \qquad \text{for every agent } i.$$

This remains true even if the allocation undergoes cycle-elimination swaps: the algorithm never assigns a more valuable good after a less valuable one so the relative order of values in $A_j$ is preserved throughout.

**Step 3: EFX verification.** Take any $g' \in A_j$. There are two possibilities:

- If $g' = g$, then the inequality from Step 1 already gives:

$$v_i(A_i) \;\geq\; v_i(A_j \setminus \{g\}).$$

- If $g' \neq g$, then $g'$ was allocated earlier, and thus $v_i(g') \geq v_i(g)$. Therefore:

$$v_i(A_j \setminus \{g'\}) = v_i(A_j) - v_i(g') \;\leq\; v_i(A_j) - v_i(g) = v_i(A_j \setminus \{g\}) \;\leq\; v_i(A_i).$$

Since the above holds for every $i$ and every $g' \in A_j$, agent $i$ never envies agent $j$ after removing any good from $A_j$. Thus $A$ is EFX.

□

### 4.1.3 EFX for Two and Three Agents

When there are very few agents, the structure of EFX allocations becomes much simpler. Specifically, an EFX allocation can always be calculated using a natural *divide-and-choose* process when there are two agents. This approach is based on the guarantee that an EFX allocation will exist when both agents evaluate bundles in the same way; the divide-and-choose protocol successfully reduces the problem to such an identical-valuation setting.

**Two Agents.** For two agents the procedure is straightforward: one agent partitions the set of goods into two bundles that they consider equally valuable and the other agent selects the bundle they prefer. Because the cutter ensures that both bundles are equal according to her own valuation, and the chooser selects her preferred bundle neither agent envies the other after the removal of any single good. Thus the resulting allocation is EFX [3, 2].

**Three Agents.** For three agents Plaut and Roughgarden [3] showed that EFX allocations can still be computed using a structured approach that generalizes divide-and-choose method combining with careful cycle-elimination steps. The key insight is that with three participants, one can repeatedly let an agent form a tentative partition and then adjust the bundles through envy-cycle elimination until no agent can envy another after the removal of any single good. This ensures that an EFX allocation always exists for $n = 3$.

**Theorem 4** (Plaut and Roughgarden, 2020)**.** *For every instance with two agents, an* EFX *allocation is guaranteed to exist.*

*Proof.* Let the two agents be 1 and 2. We begin by fixing an arbitrary agent, say agent 1, and temporarily assume that *both* agents evaluate bundles using agent 1's valuation function $v_1$.

**Step 1: Construct two EFX bundles under identical valuations.** Under the assumption that both agents share valuation $v_1$, we compute an EFX allocation $(B_1, B_2)$ of the goods between two "identical" agents.

- If valuations are additive, we may use the Envy-Cycle Elimination algorithm, which is known to return an EFX allocation under identical valuations.
- If valuations are non-additive, we may instead apply the lexicographically-maximal *leximin++* rule, which is guaranteed to produce an EFX allocation in identical-valuation settings.

Thus, $(B_1, B_2)$ is an EFX allocation according to $v_1$.

**Step 2: Let the real agent 2 choose.** We now return to the true valuations. Agent 2 is allowed to choose whichever of the two bundles she prefers:

$$B_j \in \{B_1, B_2\} \quad \text{s.t.} \quad v_2(B_j) \geq v_2(B_{3-j}).$$

Agent 2 receives her preferred bundle, and the remaining bundle is assigned to agent 1.

**Step 3: Verify EFX for both agents.**

**Agent 2.** Since agent 2 receives her favorite of the two bundles, she cannot envy agent 1 even after the removal of any single good from agent 1's bundle. Thus, agent 2 satisfies EFX.

**Agent 1.** Recall that $(B_1, B_2)$ was constructed to be EFX *with respect to* $v_1$. This means that, according to $v_1$, agent 1 does not envy bundle $B_j$ after removing any good from it. Therefore, regardless of which bundle agent 2 selects, the leftover bundle is always one toward which agent 1 has EFX satisfaction.

Since both agents satisfy the EFX condition, the resulting allocation is EFX. □

Recent developments have expanded the scope of settings in which EFX allocations can be computed efficiently. Goldberg et al. [10] established that for two-agent instances, EFX allocations remain tractable even under richer valuation classes such as *gross substitutes* and *budget-additive* valuations. However, they also proved that for *submodular* valuations, the problem is PLS-complete, indicating that no polynomial-time algorithm is likely to exist unless PLS collapses.

For the case of three agents, a significant breakthrough was achieved by Chaudhury et al. [9], who presented the first pseudo-polynomial-time algorithm guaranteeing the existence of an EFX allocation. Their proof involves intricate structural arguments and extensive case analysis. A later work by Akrami et al. [19] provided a conceptually simpler proof of the same result. Despite this progress, obtaining a *polynomial-time* algorithm for computing EFX allocations with three agents remains one of the major open problems in fair division.

| **Valuation Class** | **# Agents** | **Algorithmic Status** | **Reference** |
|---|---|---|---|
| Identical valuations | Any | Polytime (leximin++, EFX guaranteed) | [3] |
| Gross substitutes | Two | Polytime computable EFX | [10] |
| Budget-additive | Two | Polytime computable EFX | [10] |
| Submodular | Two | PLS-complete (hard) | [10] |
| General valuations | Three | Pseudo-polynomial time | [9] |
| General valuations | Three | Simpler pseudo-poly proof | [19] |
| General valuations | Three | Open (polytime) | |

Table 4.1: Comparison of valuation classes and the computational status of finding EFX allocations.

### 4.1.4 Bi-valued Valuations

In settings where each agent assigns one of only two possible values to every goodthe so-called *bi-valued* valuation modelAmanatidis et al. [16] proved that an EFX allocation always exists and, moreover, can be computed efficiently for *any* number of agents. Subsequently, Garg and Murhekar [20] strengthened this result by showing that, even under the additional requirement of *Pareto optimality (PO)*, one can still compute allocations that satisfy both EFX and PO simultaneously.

Amanatidis et al. [16] also uncovered a notable structural connection: for bi-valued instances, every *maximum Nash welfare (MNW)* allocation is guaranteed to satisfy EFX. This phenomenon extends partially beyond bi-valued settings. Babaioff et al. [24] later showed that, for general (not necessarily bi-valued) valuations with *binary marginals*that is, marginal values in $\{0, 1\}$every MNW allocation again satisfies EFX.

**Example (Bi-valued Setting).**

Consider three agents and four goods $g_1, g_2, g_3, g_4$. Each agent assigns value either "high" ($H$) or "low" ($L$) to each good. Suppose the valuation matrix is:

| | $g_1$ | $g_2$ | $g_3$ | $g_4$ |
|---|---|---|---|---|
| Agent 1 | $H$ | $H$ | $L$ | $L$ |
| Agent 2 | $H$ | $L$ | $H$ | $L$ |
| Agent 3 | $L$ | $H$ | $H$ | $L$ |

Table 4.2: Example of bi-valued valuations with $H > L$.

An MNW allocation for this instance may be:

$$A_1 = \{g_1\}, \qquad A_2 = \{g_3\}, \qquad A_3 = \{g_2, g_4\}.$$

- Agents 1 and 2 each receive a high-valued good.
- Agent 3 receives one high-valued and one low-valued good.

Since the valuations are bi-valued, removing any single good from another agent's bundle cannot raise envy: every agent sees at most one high-valued item in the others' bundles. Thus the allocation satisfies EFX, illustrating the structural guarantee proved in [16, 24].

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